BE (6th Semester)
Examination, April-May, 2016

(Old Scheme)

Advanced Concrete Technology

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Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 28

Note : (*i*) Attempt part (*a*) of all units, and any **two** parts from (*b*), (*c*) and (*d*).

(*ii*) Assume suitable value with justification if necessary.

(*iii*) Answers should be supported by suitable figures wherever necessary.

(*iv*) The figures in the right-hand margin indicate marks.

Unit - I

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1. (*a*) What do you mean by aspect ratio related to fibers? 2

(*b*) What is classification of aggregate? Explain any three of them. 7

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(*c*) What is hydration? Describe the structure of hydrated cement. How does hydration process in PPC differ than OPC? 7

(*d*) Write short notes on (any **two**) : 7

(*i*) Alkali aggregate reaction

(*ii*) Gradation of aggregate

(*iii*) Bulking of aggregate

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Unit - II

2. (*a*) What is workability of concrete? 2

(*b*) What is Admixture? Describe any two commonly used admixtures. 7

(*c*) Explain the following terms related to concrete and their internal relation : 7

(*i*) Strength

(*ii*) Creep

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(*iii*) Modulus of elasticity

(*d*) Write short notes on (any **two**) : 7

(*i*) Shrinkage of concrete

(*ii*) Maturity of concrete

(*iii*) Accelerated curing test

Unit - III

3. (*a*) What is mix proportioning? 2

(*b*) Differentiate the salient features of ACI method and IS-Code method of concrete mix proportioning. 7

(*c*) What are the basics of mix design? 7

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(*d*) Explain the following related to concrete mix design : 7

(*i*) Target mean strength and w/c ratio

(*ii*) Testing of concrete

(*iii*) Acceptance criteria

Unit - IV

4. (*a*) Explain No fines concrete. 2

(*b*) What is polymer concrete? Explain their classification in detail. 7

(*c*) What is Light Weight Concrete? Explain their requirement / suitability. 7

(*d*) Write short notes on (any two) : 7

(*i*) Fiber reinforced concrete

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(*ii*) Cold weather concrete

(*iii*) Precautions for hot weather concrete

Unit - V

5. (*a*) What are the methods of transportation of concrete? 2

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(*b*) Describe vacum dewatered concrete with suitable diagrams. 7

(*c*) Explain the necessity and process of RMC. 7

(*d*) Write short notes on (any two) : 7

(*i*) Guniting

(*ii*) Ferro cement

(*iii*) Underwater concreting